

class12

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```
counts <-read.csv("airway_scaledcounts.csv", row.names=1)
metadata <- read.csv("airway_metadata.csv")
```

```
head(counts)
```

	SRR1039508	SRR1039509	SRR1039512	SRR1039513	SRR1039516
ENSG00000000003	723	486	904	445	1170
ENSG00000000005	0	0	0	0	0
ENSG000000000419	467	523	616	371	582
ENSG000000000457	347	258	364	237	318
ENSG000000000460	96	81	73	66	118
ENSG000000000938	0	0	1	0	2

	SRR1039517	SRR1039520	SRR1039521
ENSG00000000003	1097	806	604
ENSG00000000005	0	0	0
ENSG000000000419	781	417	509
ENSG000000000457	447	330	324
ENSG000000000460	94	102	74
ENSG000000000938	0	0	0

```
head(metadata)
```

	id	dex	celltype	geo_id
1	SRR1039508	control	N61311	GSM1275862
2	SRR1039509	treated	N61311	GSM1275863
3	SRR1039512	control	N052611	GSM1275866
4	SRR1039513	treated	N052611	GSM1275867
5	SRR1039516	control	N080611	GSM1275870
6	SRR1039517	treated	N080611	GSM1275871

```
nrow(counts)
```

```
[1] 38694
```

```
nrow(metadata)
```

```
[1] 8
```

Q1. Based on data above, there are 38694 genes in the dataset.

```
View(metadata)
```

##Q2. After viewing the metadata, there are 4 control cell lines.

```
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
control <- metadata %>% filter(dex=="control")
control.counts <- counts %>% select(control$id)
control.mean <- rowSums(control.counts)/4
head(control.mean)
```

```
ENSG000000000003 ENSG000000000005 ENSG000000000419 ENSG000000000457 ENSG000000000460
          900.75           0.00           520.50           339.75           97.25
ENSG0000000000938
          0.75
```

I like the dplyr package because it makes sense in my head to “filter” for different variables.

Q3. You could assign the 4 values to a vector that is the actual number of controls in case the value were to change.

##Q4. shown below.

```
library(dplyr)

treated <- metadata %>% filter(dex == "treated")

treated.counts <- counts %>% select(treated$id)

treated.mean <- rowMeans(treated.counts)

head(treated.mean)
```

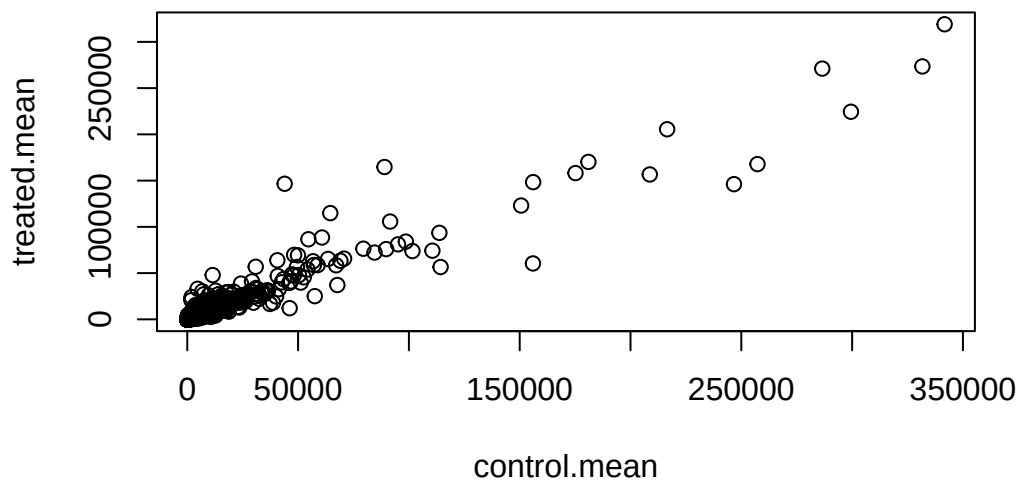
```
ENSG00000000003 ENSG00000000005 ENSG00000000419 ENSG00000000457 ENSG00000000460
      658.00           0.00          546.00          316.50          78.75
ENSG000000000938
      0.00
```

```
#combining meancount data

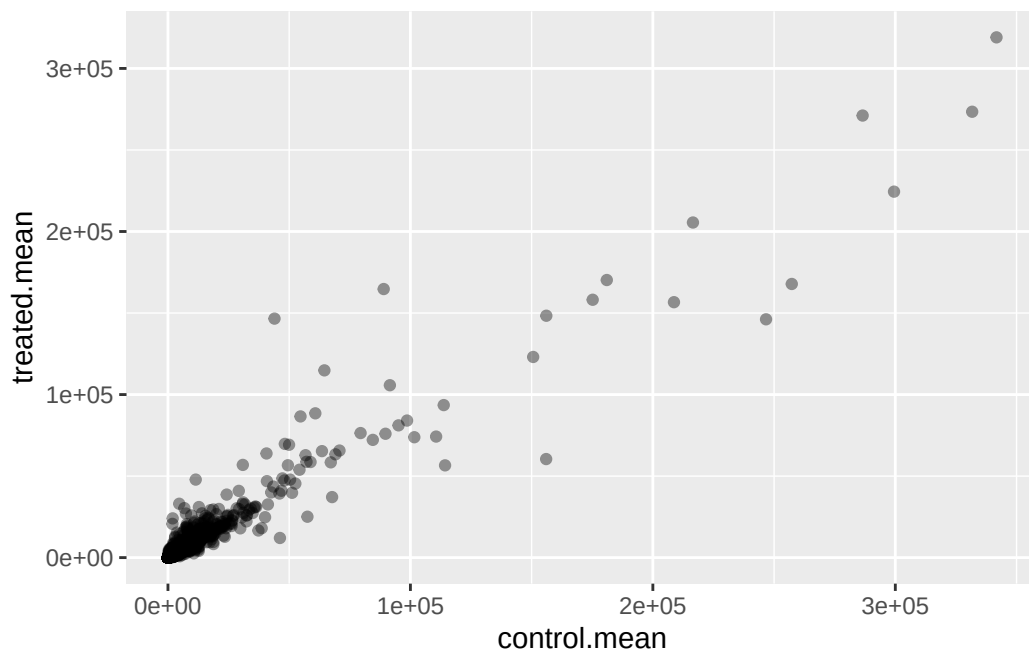
meancounts <- data.frame(control.mean, treated.mean)
```

Q5a. shown below.

```
plot(meancounts)
```



```
library(ggplot2)
ggplot(meancounts, aes(control.mean, treated.mean))+
  geom_point(alpha=.4)
```



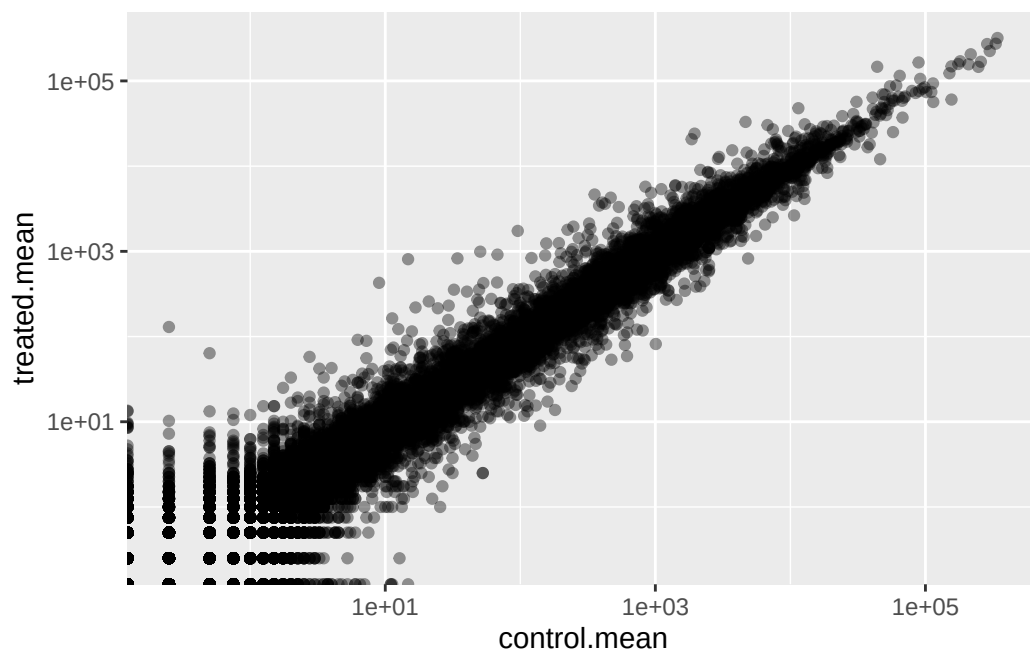
```
#alpha used to change opacity
```

Q5b. shown above. Using “geom_point” in the ggplot2 package.

```
library(ggplot2)
ggplot(meancounts, aes(control.mean, treated.mean))+
  geom_point(alpha=.4)+
  scale_x_log10()+
  scale_y_log10()
```

Warning in scale_x_log10(): log-10 transformation introduced infinite values.

Warning in scale_y_log10(): log-10 transformation introduced infinite values.

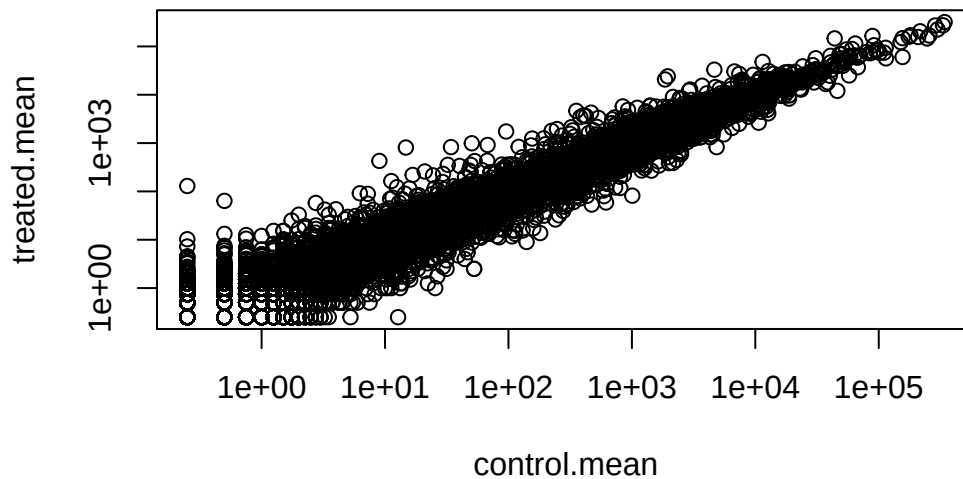


```
#scaled using ggplot
```

```
plot(meancounts, log="xy")
```

Warning in xy.coords(x, y, xlabel, ylabel, log): 15032 x values <= 0 omitted from logarithmic plot

Warning in xy.coords(x, y, xlabel, ylabel, log): 15281 y values <= 0 omitted from logarithmic plot



Q6. shown above with ggplot and default plot().

```
meancounts$log2fc <- log2(meancounts[, "treated.mean"] / meancounts[, "control.mean"])
head(meancounts)
```

	control.mean	treated.mean	log2fc
ENSG000000000003	900.75	658.00	-0.45303916
ENSG000000000005	0.00	0.00	NaN
ENSG000000000419	520.50	546.00	0.06900279
ENSG000000000457	339.75	316.50	-0.10226805
ENSG000000000460	97.25	78.75	-0.30441833
ENSG000000000938	0.75	0.00	-Inf

```
#filtered data
zero.vals <- which(meancounts[,1:2]==0, arr.ind=TRUE)

to.rm <- unique(zero.vals[,1])
mycounts <- meancounts[-to.rm,]
head(mycounts)
```

	control.mean	treated.mean	log2fc
ENSG000000000003	900.75	658.00	-0.45303916
ENSG000000000419	520.50	546.00	0.06900279
ENSG000000000457	339.75	316.50	-0.10226805
ENSG000000000460	97.25	78.75	-0.30441833
ENSG000000000971	5219.00	6687.50	0.35769358
ENSG000000001036	2327.00	1785.75	-0.38194109

Q7. arr.ind = TRUE makes which() return the row and column positions of all TRUE values instead of just a single vector of positions, and gives rows and columns that have zeroes. Then using unique() so that if a gene has zeros in multiple samples it's only listed once.

```
up.ind <- mycounts$log2fc > 2
down.ind <- mycounts$log2fc < (-2)
```

```
table(up.ind)
```

```
up.ind
FALSE TRUE
21567  250
```

```
table(down.ind)
```

```
down.ind
FALSE TRUE
21450  367
```

Q8. Using results above, there are 250 up regulated changes at the greater than 2 fc level.

Q9. From the table above, there are 367 down regulated genes at the greater than 2 fc level.

Q10. The results could be misleading because it's based on fold-changes, which could be large without being statistically significant.

```
library(DESeq2)
```

```
Loading required package: S4Vectors
```

```
Loading required package: stats4
```

```
Loading required package: BiocGenerics
```

```
Loading required package: generics
```

```
Attaching package: 'generics'
```

```
The following object is masked from 'package:dplyr':
```

```
  explain
```

```
The following objects are masked from 'package:base':
```

```
  as.difftime, as.factor, as.ordered, intersect, is.element, setdiff,  
  setequal, union
```

```
Attaching package: 'BiocGenerics'
```

```
The following object is masked from 'package:dplyr':
```

```
  combine
```


The following objects are masked from 'package:stats':

IQR, mad, sd, var, xtabs

The following objects are masked from 'package:base':

anyDuplicated, aperm, append, as.data.frame, basename, cbind,
colnames, dirname, do.call, duplicated, eval, evalq, Filter, Find,
get, grep, grepl, is.unsorted, lapply, Map, mapply, match, mget,
order, paste, pmax, pmax.int, pmin, pmin.int, Position, rank,
rbind, Reduce, rownames, sapply, saveRDS, table, tapply, unique,
unsplit, which.max, which.min

Attaching package: 'S4Vectors'

The following objects are masked from 'package:dplyr':

first, rename

The following object is masked from 'package:utils':

findMatches

The following objects are masked from 'package:base':

expand.grid, I, unname

Loading required package: IRanges

Attaching package: 'IRanges'

The following objects are masked from 'package:dplyr':

collapse, desc, slice

Loading required package: GenomicRanges

Loading required package: Seqinfo

Loading required package: SummarizedExperiment

Loading required package: MatrixGenerics

Loading required package: matrixStats

Attaching package: 'matrixStats'

The following object is masked from 'package:dplyr':

count

Attaching package: 'MatrixGenerics'

The following objects are masked from 'package:matrixStats':

colAlls, colAnyNAs, colAnys, colAveragesPerRowSet, colCollapse,
colCounts, colCummaxs, colCummins, colCumprods, colCumsums,
colDiffs, colIQRDiffs, colIQRs, colLogSumExps, colMadDiffs,
colMads, colMaxs, colMeans2, colMedians, colMins, colOrderStats,
colProds, colQuantiles, colRanges, colRanks, colSdDiffs, colSds,
colSums2, colTabulates, colVarDiffs, colVars, colWeightedMads,
colWeightedMeans, colWeightedMedians, colWeightedSds,
colWeightedVars, rowAlls, rowAnyNAs, rowAnys, rowAveragesPerColSet,
rowCollapse, rowCounts, rowCummaxs, rowCummins, rowCumprods,
rowCumsums, rowDiffs, rowIQRDiffs, rowIQRs, rowLogSumExps,
rowMadDiffs, rowMads, rowMaxs, rowMeans2, rowMedians, rowMins,
rowOrderStats, rowProds, rowQuantiles, rowRanges, rowRanks,
rowSdDiffs, rowSds, rowSums2, rowTabulates, rowVarDiffs, rowVars,
rowWeightedMads, rowWeightedMeans, rowWeightedMedians,
rowWeightedSds, rowWeightedVars

Loading required package: Biobase

Welcome to Bioconductor

Vignettes contain introductory material; view with
'browseVignettes()'. To cite Bioconductor, see
'citation("Biobase")', and for packages 'citation("pkgname")'.

Attaching package: 'Biobase'

The following object is masked from 'package:MatrixGenerics':

rowMedians

The following objects are masked from 'package:matrixStats':

anyMissing, rowMedians

```
citation("DESeq2")
```

To cite package 'DESeq2' in publications use:

Love, M.I., Huber, W., Anders, S. Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2 Genome Biology 15(12):550 (2014)

A BibTeX entry for LaTeX users is

```
@Article{,
  title = {Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2},
  author = {Michael I. Love and Wolfgang Huber and Simon Anders},
  year = {2014},
  journal = {Genome Biology},
  doi = {10.1186/s13059-014-0550-8},
  volume = {15},
  issue = {12},
  pages = {550},
}
```

```
dds <- DESeqDataSetFromMatrix(countData=counts,
                              colData=metadata,
                              design=~dex)
```

converting counts to integer mode

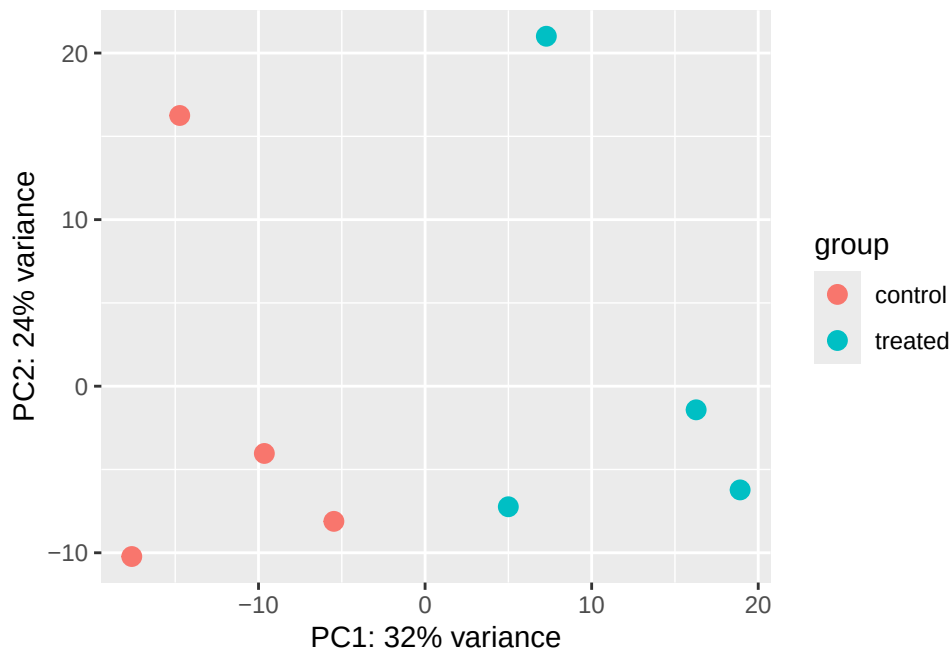
Warning in DESeqDataSet(se, design = design, ignoreRank): some variables in design formula are characters, converting to factors

```
dds
```

```
class: DESeqDataSet
dim: 38694 8
metadata(1): version
assays(1): counts
rownames(38694): ENSG000000000003 ENSG000000000005 ... ENSG00000283120
               ENSG00000283123
rowData names(0):
colnames(8): SRR1039508 SRR1039509 ... SRR1039520 SRR1039521
colData names(4): id dex celltype geo_id
```

```
vsd <- vst(dds, blind = FALSE)
plotPCA(vsd, intgroup = c("dex"))
```

using ntop=500 top features by variance



```
pcaData <- plotPCA(vsd, intgroup=c("dex"), returnData=TRUE)
```

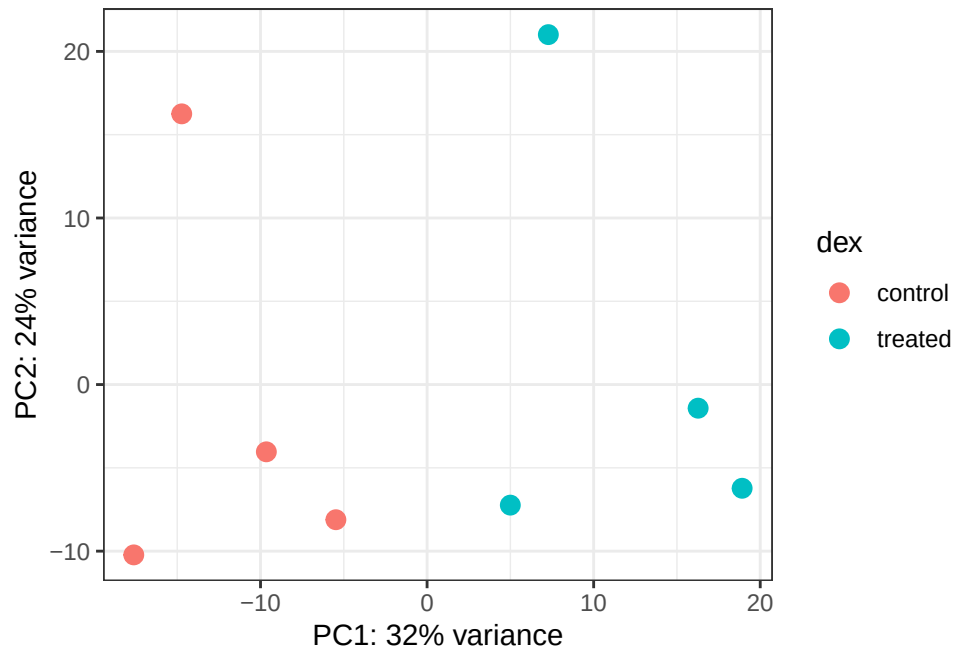
using ntop=500 top features by variance

```
head(pcaData)
```

	PC1	PC2	group	name	id	dex	celltype
SRR1039508	-17.607922	-10.225252	control	SRR1039508	SRR1039508	control	N61311
SRR1039509	4.996738	-7.238117	treated	SRR1039509	SRR1039509	treated	N61311
SRR1039512	-5.474456	-8.113993	control	SRR1039512	SRR1039512	control	N052611
SRR1039513	18.912974	-6.226041	treated	SRR1039513	SRR1039513	treated	N052611
SRR1039516	-14.729173	16.252000	control	SRR1039516	SRR1039516	control	N080611
SRR1039517	7.279863	21.008034	treated	SRR1039517	SRR1039517	treated	N080611
	geo_id	sizeFactor					
SRR1039508	GSM1275862	1.0193796					
SRR1039509	GSM1275863	0.9005653					
SRR1039512	GSM1275866	1.1784239					
SRR1039513	GSM1275867	0.6709854					
SRR1039516	GSM1275870	1.1731984					
SRR1039517	GSM1275871	1.3929361					

```
percentVar <- round(100 * attr(pcaData, "percentVar"))
```

```
ggplot(pcaData) +  
  aes(x = PC1, y = PC2, color = dex) +  
  geom_point(size = 3) +  
  xlab(paste0("PC1: ", percentVar[1], "% variance")) +  
  ylab(paste0("PC2: ", percentVar[2], "% variance")) +  
  coord_fixed() +  
  theme_bw()
```



```
dds <- DESeq(dds)
```

```
estimating size factors
```

```
estimating dispersions
```

```
gene-wise dispersion estimates
```

```
mean-dispersion relationship
```

```
final dispersion estimates
```

```
fitting model and testing
```

```
# results
```

```
res <- results(dds)
```

```
res
```

log2 fold change (MLE): dex treated vs control

Wald test p-value: dex treated vs control

DataFrame with 38694 rows and 6 columns

	baseMean	log2FoldChange	lfcSE	stat	pvalue
	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>
ENSG000000000003	747.1942	-0.350703	0.168242	-2.084514	0.0371134
ENSG000000000005	0.0000	NA	NA	NA	NA
ENSG000000000419	520.1342	0.206107	0.101042	2.039828	0.0413675
ENSG000000000457	322.6648	0.024527	0.145134	0.168996	0.8658000
ENSG000000000460	87.6826	-0.147143	0.256995	-0.572550	0.5669497
...	...	...	...	...	...
ENSG00000283115	0.000000	NA	NA	NA	NA
ENSG00000283116	0.000000	NA	NA	NA	NA
ENSG00000283119	0.000000	NA	NA	NA	NA
ENSG00000283120	0.974916	-0.66825	1.69441	-0.394385	0.693297
ENSG00000283123	0.000000	NA	NA	NA	NA
	padj				
	<numeric>				
ENSG000000000003	0.163017				
ENSG000000000005	NA				
ENSG000000000419	0.175937				
ENSG000000000457	0.961682				
ENSG000000000460	0.815805				
...	...				
ENSG00000283115	NA				
ENSG00000283116	NA				
ENSG00000283119	NA				
ENSG00000283120	NA				
ENSG00000283123	NA				

```
summary(res)
```

out of 25258 with nonzero total read count

adjusted p-value < 0.1

LFC > 0 (up) : 1564, 6.2%

LFC < 0 (down) : 1188, 4.7%

outliers [1] : 142, 0.56%

low counts [2] : 9971, 39%

(mean count < 10)

[1] see 'cooksCutoff' argument of ?results

[2] see 'independentFiltering' argument of ?results

```
library("AnnotationDbi")
```

Attaching package: 'AnnotationDbi'

The following object is masked from 'package:dplyr':

select

```
library("org.Hs.eg.db")
```

```
columns(org.Hs.eg.db)
```

[1]	"ACCNUM"	"ALIAS"	"ENSEMBL"	"ENSEMBLPROT"	"ENSEMBLTRANS"
[6]	"ENTREZID"	"ENZYME"	"EVIDENCE"	"EVIDENCEALL"	"GENENAME"
[11]	"GENETYPE"	"GO"	"GOALL"	"IPI"	"MAP"
[16]	"OMIM"	"ONTOLOGY"	"ONTOLOGYALL"	"PATH"	"PFAM"
[21]	"PMID"	"PROSITE"	"REFSEQ"	"SYMBOL"	"UCSCKG"
[26]	"UNIPROT"				

```
res$symbol <- mapIds(org.Hs.eg.db,  
  keys=row.names(res), # genenames  
  keytype="ENSEMBL",   #format of our genenames  
  column="SYMBOL",     # The new format  
  multiVals="first")
```

'select()' returned 1:many mapping between keys and columns

```
head(res)
```

log2 fold change (MLE): dex treated vs control

Wald test p-value: dex treated vs control

DataFrame with 6 rows and 7 columns

	baseMean	log2FoldChange	lfcSE	stat	pvalue
	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>
ENSG00000000003	747.194195	-0.350703	0.168242	-2.084514	0.0371134

ENSG000000000005	0.000000	NA	NA	NA	NA
ENSG0000000000419	520.134160	0.206107	0.101042	2.039828	0.0413675
ENSG0000000000457	322.664844	0.024527	0.145134	0.168996	0.8658000
ENSG0000000000460	87.682625	-0.147143	0.256995	-0.572550	0.5669497
ENSG0000000000938	0.319167	-1.732289	3.493601	-0.495846	0.6200029
	padj	symbol			
	<numeric>	<character>			
ENSG0000000000003	0.163017	TSPAN6			
ENSG0000000000005	NA	TNMD			
ENSG0000000000419	0.175937	DPM1			
ENSG0000000000457	0.961682	SCYL3			
ENSG0000000000460	0.815805	FIRRM			
ENSG0000000000938	NA	FGR			

Q11. Shown below.

```
res$entrez <- mapIds(org.Hs.eg.db,
                     keys=row.names(res),
                     column="ENTREZID",
                     keytype="ENSEMBL",
                     multiVals="first")
```

'select()' returned 1:many mapping between keys and columns

```
res$uniprot <- mapIds(org.Hs.eg.db,
                     keys=row.names(res),
                     column="UNIPROT",
                     keytype="ENSEMBL",
                     multiVals="first")
```

'select()' returned 1:many mapping between keys and columns

```
res$genename <- mapIds(org.Hs.eg.db,
                      keys=row.names(res),
                      column="GENENAME",
                      keytype="ENSEMBL",
                      multiVals="first")
```

'select()' returned 1:many mapping between keys and columns

```
head(res)
```

log2 fold change (MLE): dex treated vs control

Wald test p-value: dex treated vs control

DataFrame with 6 rows and 10 columns

	baseMean	log2FoldChange	lfcSE	stat	pvalue
	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>
ENSG000000000003	747.194195	-0.350703	0.168242	-2.084514	0.0371134
ENSG000000000005	0.000000	NA	NA	NA	NA
ENSG000000000419	520.134160	0.206107	0.101042	2.039828	0.0413675
ENSG000000000457	322.664844	0.024527	0.145134	0.168996	0.8658000
ENSG000000000460	87.682625	-0.147143	0.256995	-0.572550	0.5669497
ENSG000000000938	0.319167	-1.732289	3.493601	-0.495846	0.6200029
	padj	symbol	entrez	uniprot	
	<numeric>	<character>	<character>	<character>	
ENSG000000000003	0.163017	TSPAN6	7105	AOA087WYV6	
ENSG000000000005	NA	TNMD	64102	Q9H2S6	
ENSG000000000419	0.175937	DPM1	8813	H0Y368	
ENSG000000000457	0.961682	SCYL3	57147	X6RHX1	
ENSG000000000460	0.815805	FIRRM	55732	A6NFP1	
ENSG000000000938	NA	FGR	2268	B7Z6W7	
		genename			
		<character>			
ENSG000000000003		tetraspanin 6			
ENSG000000000005		tenomodulin			
ENSG000000000419		dolichyl-phosphate m..			
ENSG000000000457		SCY1 like pseudokina..			
ENSG000000000460		FIGNL1 interacting r..			
ENSG000000000938		FGR proto-oncogene, ..			

```
# viewing by p-value
```

```
ord <- order( res$padj )
#View(res[ord,])
head(res[ord,])
```

log2 fold change (MLE): dex treated vs control

Wald test p-value: dex treated vs control

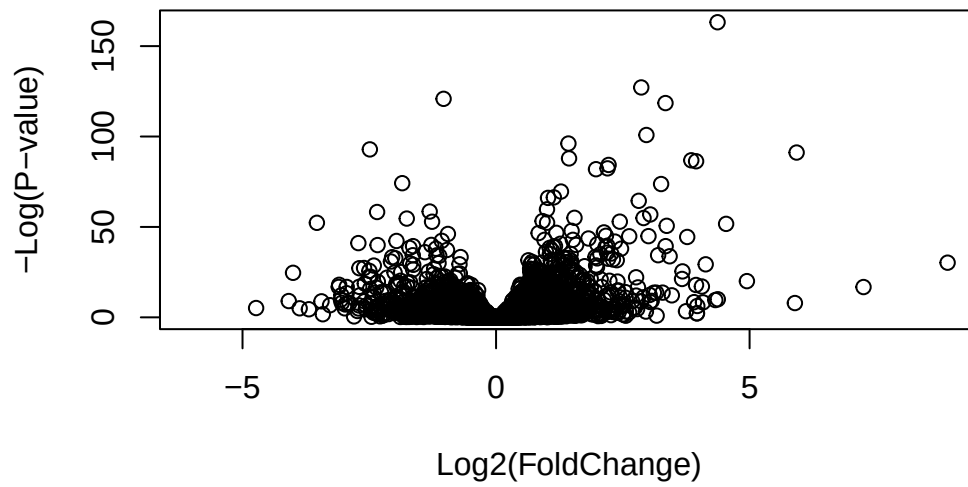
DataFrame with 6 rows and 10 columns

	baseMean	log2FoldChange	lfcSE	stat	pvalue
	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>

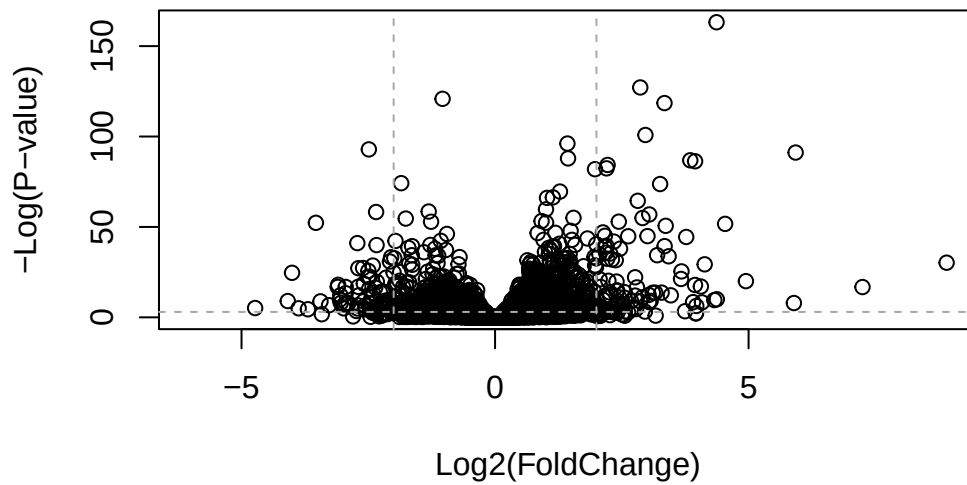
ENSG00000152583	954.771	4.36836	0.2371306	18.4217	8.79214e-76
ENSG00000179094	743.253	2.86389	0.1755659	16.3123	8.06568e-60
ENSG00000116584	2277.913	-1.03470	0.0650826	-15.8983	6.51317e-57
ENSG00000189221	2383.754	3.34154	0.2124091	15.7316	9.17960e-56
ENSG00000120129	3440.704	2.96521	0.2036978	14.5569	5.27883e-48
ENSG00000148175	13493.920	1.42717	0.1003811	14.2175	7.13625e-46
	padj	symbol	entrez	uniprot	
	<numeric>	<character>	<character>	<character>	
ENSG00000152583	1.33157e-71	SPARCL1	8404	B4E2Z0	
ENSG00000179094	6.10774e-56	PER1	5187	A2I2P6	
ENSG00000116584	3.28806e-53	ARHGEF2	9181	A0A8Q3SIN5	
ENSG00000189221	3.47563e-52	MAOA	4128	B4DF46	
ENSG00000120129	1.59896e-44	DUSP1	1843	B4DRR4	
ENSG00000148175	1.80131e-42	STOM	2040	F8VSL7	
	genename				
	<character>				
ENSG00000152583	SPARC like 1				
ENSG00000179094	period circadian reg..				
ENSG00000116584	Rho/Rac guanine nucl..				
ENSG00000189221	monoamine oxidase A				
ENSG00000120129	dual specificity pho..				
ENSG00000148175	stomatin				

```
write.csv(res[ord,], "deseq_results.csv")
```

```
plot( res$log2FoldChange, -log(res$padj),
      xlab="Log2(FoldChange)",
      ylab="-Log(P-value)")
```



```
plot( res$log2FoldChange, -log(res$padj),  
      ylab="-Log(P-value)", xlab="Log2(FoldChange)")  
  
# Add some cut-off lines  
abline(v=c(-2,2), col="darkgray", lty=2)  
abline(h=-log(0.05), col="darkgray", lty=2)
```

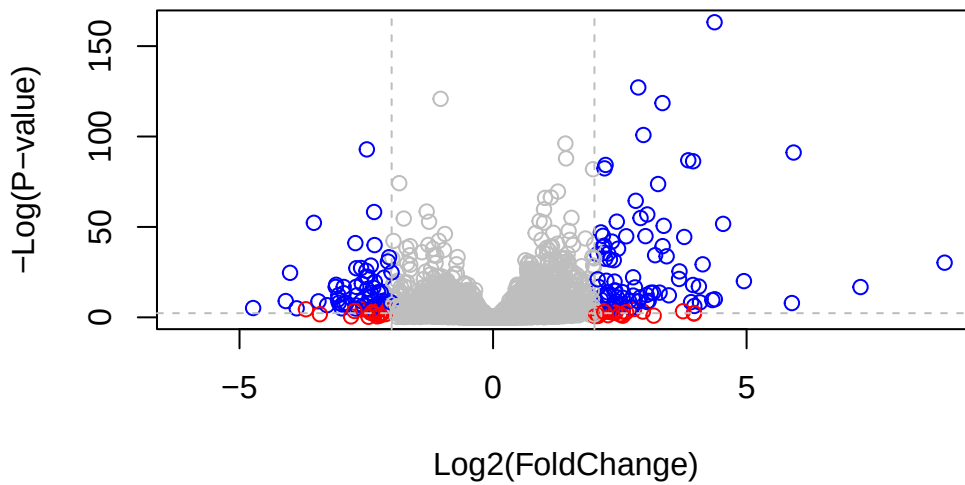


```
mycols <- rep("gray", nrow(res))
mycols[ abs(res$log2FoldChange) > 2 ] <- "red"

inds <- (res$padj < 0.01) & (abs(res$log2FoldChange) > 2 )
mycols[ inds ] <- "blue"

plot( res$log2FoldChange, -log(res$padj),
      col=mycols, ylab="-Log(P-value)", xlab="Log2(FoldChange)" )

abline(v=c(-2,2), col="gray", lty=2)
abline(h=-log(0.1), col="gray", lty=2)
```



```
library(EnhancedVolcano)
```

Loading required package: ggrepel

```
x <- as.data.frame(res)
```

```
EnhancedVolcano(x,
  lab = x$symbol,
  x = 'log2FoldChange',
  y = 'pvalue')
```

Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
 i Please use `linewidth` instead.
 i The deprecated feature was likely used in the EnhancedVolcano package.
 Please report the issue to the authors.

Warning: The `size` argument of `element\_line()` is deprecated as of ggplot2 3.4.0.
 i Please use the `linewidth` argument instead.
 i The deprecated feature was likely used in the EnhancedVolcano package.
 Please report the issue to the authors.

Volcano plot

EnhancedVolcano

